Corso: Data Science and Database Technology
Quiz: Exam 2026-01-29

PORTALE ESAMI

[redacted]

Iniziato	29 gennaio 2026, 14:11
Stato	Completato
Terminato	29 gennaio 2026, 15:51
Tempo impiegato	1 ora 40 min.
Valutazione	**31,75** su un massimo di 32,00 (**99**%)
Riepilogo del tentativo	1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Domanda 1
Risposta corretta
Punteggio ottenuto 1,60 su 1,60

Conceptual schema (1.6 points, -15% penalty for each wrong answer)

We want to analyze the information relating to failure management for traffic lights located in urban areas in italian cities.

The location of the traffic light where the failure occurred is known, including the neighborhood, district, city, province, and region, as well as whether it was located at a high-traffic intersection. The list of facilities available in each neighborhood such as schools, hospitals, children's play areas, etc, is also provided. Also the time since traffic light was installed is known (one value between less than 1 year, between 1 and 5 years, between 6 and 10 years, more than 10 years).

Failures are characterized by their causes (one or more values between 'Electrical', 'Environmental', 'Mechanical', 'External', 'Operational') and the indication of their severity level (one value between 'Low', 'Medium' and 'High'). Also the required type of intervention is available (one or more values between 'Repair', 'Replacement', 'Software Update'). Companies involved in the repair of the failures are characterized by their geographical location (expressed in terms of city, region and country) and their size (one value between small, medium, large).

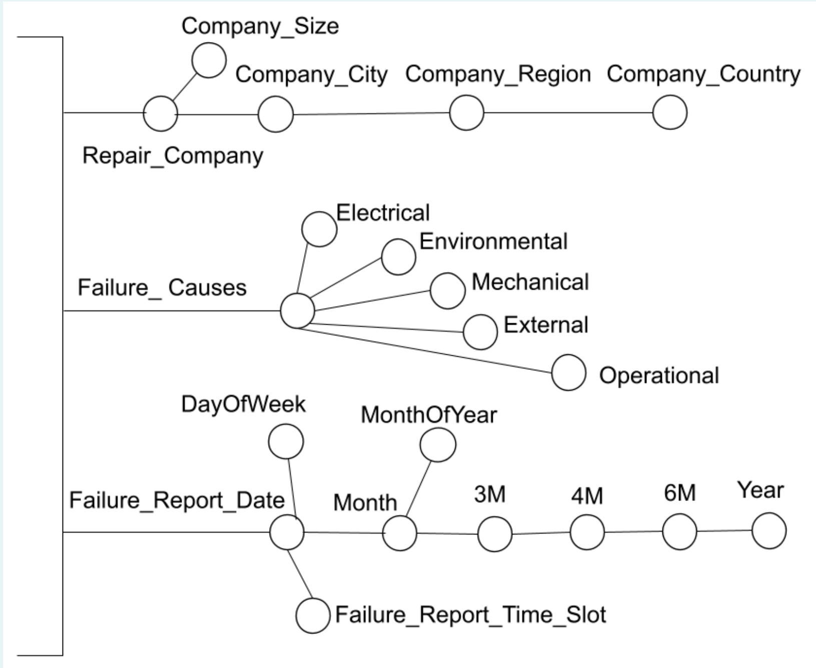
You want to store the time the failure report was received, expressed in terms of date, day of the week, month, month of the year, quarter (3M), four-month period (4M), six-month period (6M), and year. Also the time slot of the failure report is known (one value between 'morning', 'afternoon', 'evening', 'night').

We want to analyze the average repair cost per traffic light and average repair duration per traffic light based on the following information:

- Neighborhood, district, city, province, and region where a traffic light failure occurred. The list of facilities (e.g., schools, hospitals, children's play areas, etc.) available in each neighborhood is also provided.
- Indication of whether the failure occurred in a traffic light located at a high-traffic intersection (one value between 'yes' and 'no')
- Time since traffic light installation (one value between less than 1 year, between 1 and 5 years, between 6 and 10 years, more than 10 years)
- Indication of failure causes (one or more values between 'Electrical', 'Environmental', 'Mechanical', 'External', 'Operational')
- Indication of the failure severity level (one value between 'Low', 'Medium' and 'High')
- Indication of the required type of intervention (one or more value between 'Repair', 'Replacement', 'Software Update')
- Repair company, with indication of the geographical location expressed in terms of city, region and country, is known. The size of the repair company is also known (one value between small, medium, large)
- Date, month, quarter (3M), four-month period (4M), six-month period (6M) and year in which the failure report was received. For the date, the indication of the day of week is also known, and for the month the indication of the month of the year is available. Finally, the time slot of the failure report is known (one value between 'morning', 'afternoon', 'evening', 'night').

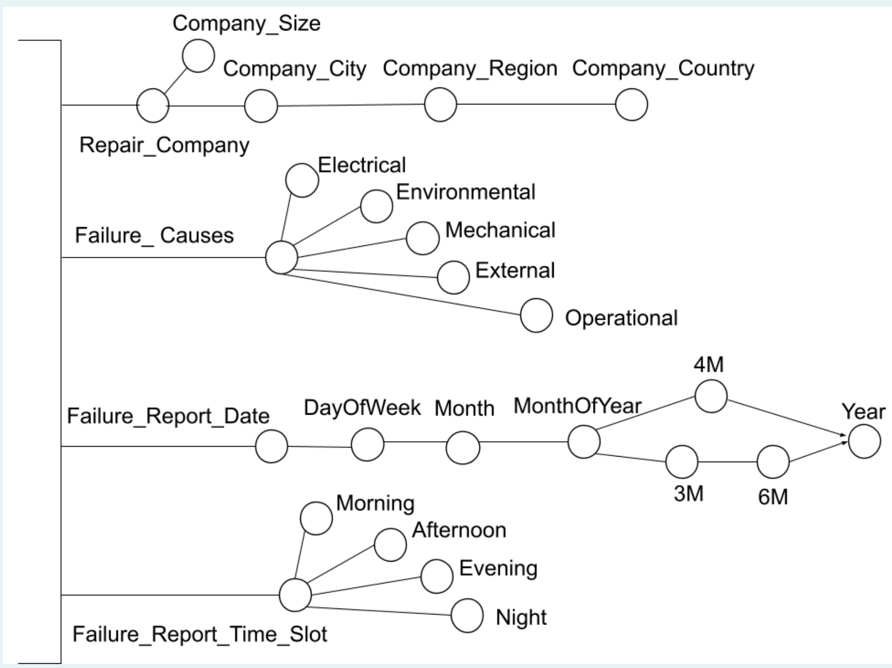
Select, from the dimensions proposed below, those that meet the requirements described in the problem specifications.

a.



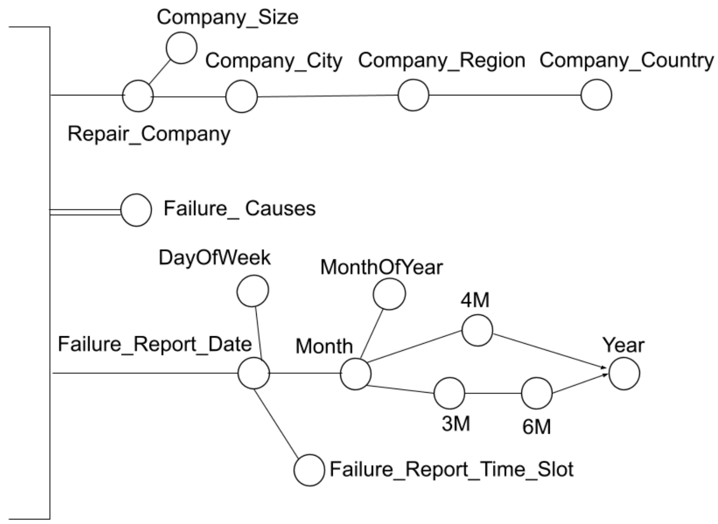
Company_Size
Company_City
Company_Region
Company_Country
Repair_Company
Electrical
Environmental
Failure_ Causes
Mechanical
External
Operational
DayOfWeek
MonthOfYear
Failure_Report_Date
Month
3M
4M
6M
Year
Failure_Report_Time_Slot

b.

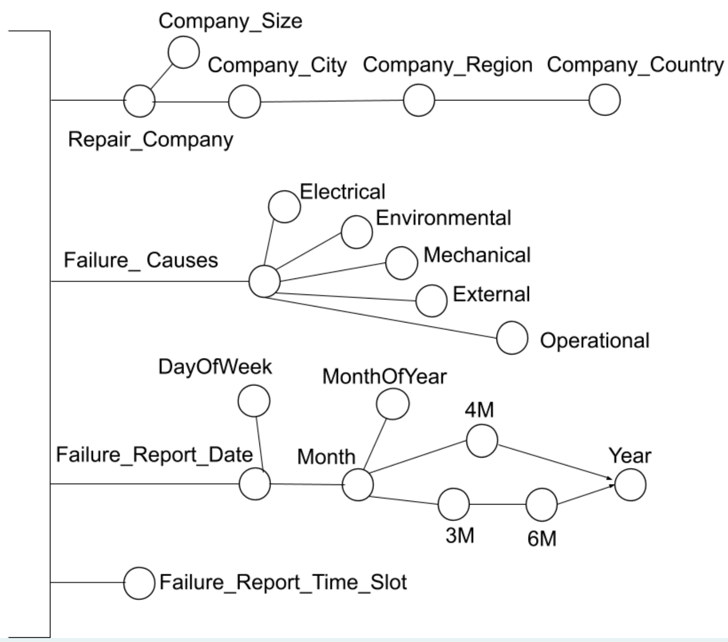


Company_Size
Company_City
Company_Region
Company_Country
Repair_Company
Electrical
Environmental
Failure_ Causes
Mechanical
External
Operational
4M
Failure_Report_Date
DayOfWeek
Month
MonthOfYear
Year
3M
6M
Morning
Afternoon
Evening
Night
Failure_Report_Time_Slot

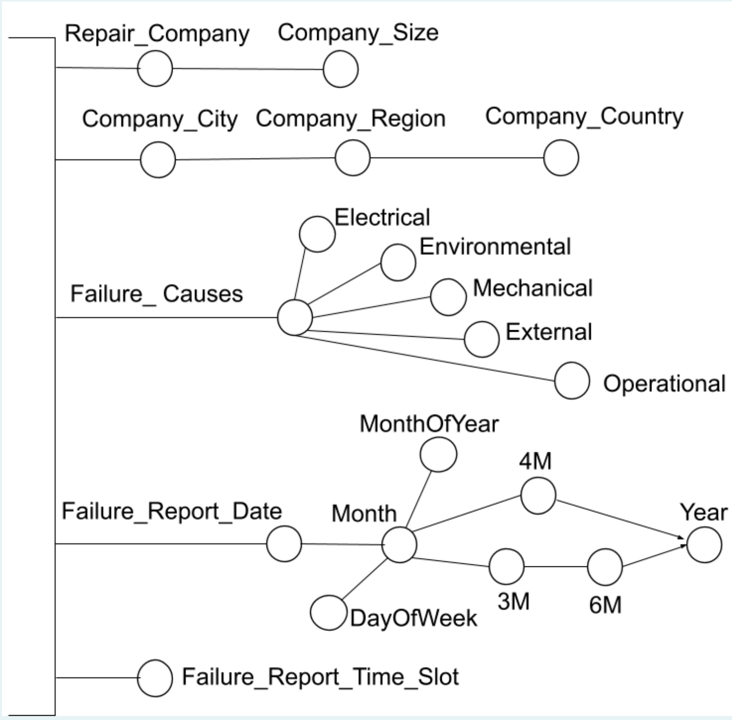
c.


Company_Size
Company_City
Company_Region
Company_Country
Repair_Company
Failure_ Causes
DayOfWeek
MonthOfYear
4M
Failure_Report_Date
Month
Year
3M
6M
Failure_Report_Time_Slot

d.

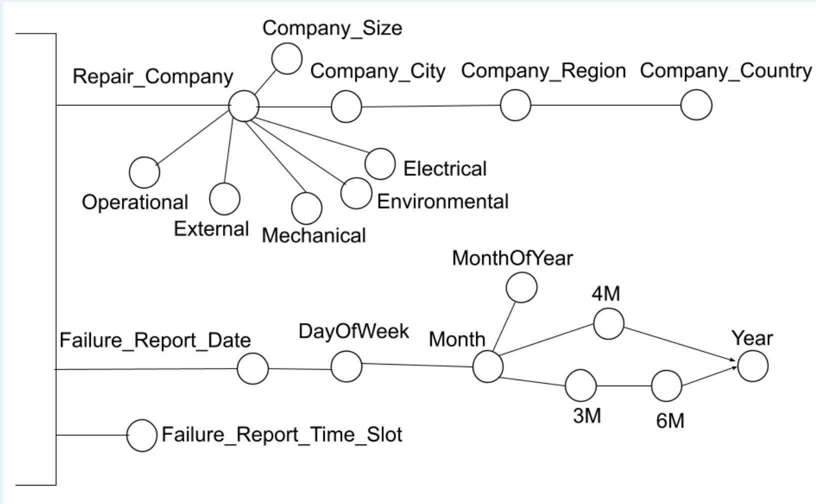

Company_Size
Company_City
Company_Region
Company_Country
Repair_Company
Electrical
Environmental
Failure_ Causes
Mechanical
External
Operational
DayOfWeek
MonthOfYear
4M
Failure_Report_Date
Month
Year
3M
6M
Failure_Report_Time_Slot

e.

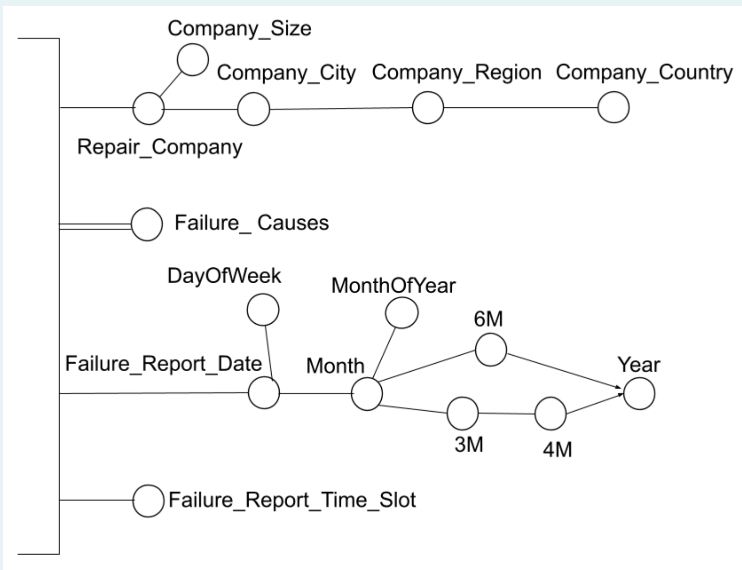

Repair_Company
Company_Size
Company_City
Company_Region
Company_Country
Electrical
Environmental
Failure_ Causes
Mechanical
External
Operational
MonthOfYear
4M
Failure_Report_Date
Month
Year
DayOfWeek
3M
6M
Failure_Report_Time_Slot

○ f. None of the other answers is correct

○ g.

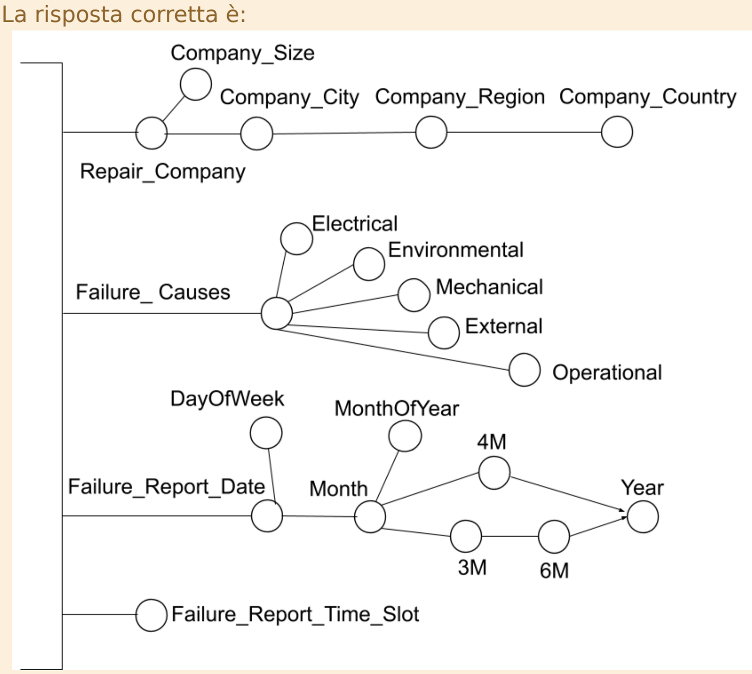


○ h.



Risposta corretta.

La risposta corretta è:



Domanda **2**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

1 point (-15% penalty for each incorrect answer)

Given the following set of points in a two dimensional space:

- P1 = (0, 1)
- P2 = (7, 0)
- P3 = (2, 0)
- P4 = (8, 1)

We apply the agglomerative hierarchical clustering using the Manhattan distance and the single-linkage (or MIN) policy.

The Manhattan distance between two points x and y is defined as $dist(x, y) = \sum_i |x_i - y_i|$.

The single-linkage (or MIN) policy between two points/clusters XXX and YYY is defined as $dist(X, Y) = min(dist(x, y))$), where x and y are points in clusters X and Y, respectively.

Which pair of points is merged first by the algorithm?

- a. None of the other answers is correct
- b. {P2, P4} ✔
- c. {P3, P4}
- d. {P1, P3}
- e. {P1, P2}
- f. {P3, P4}

Risposta corretta.

La risposta corretta è:
{P2, P4}

Domanda **3**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Cardinalities (1 point, -15% penalty for each wrong answer)

The following tables are provided:

```
BOOK (BCode, Title, ACode, PCode, Category, PublicationYear, Price)
AUTHOR (ACode, FirstName, LastName, Country, BirthDate)
PUBLISHER (PCode, PName, FoundationDate, HeadquartersRegion)
CUSTOMER (CCode, FirstName, LastName, BirthDate, ResidenceProvince, LoyaltyLevel)
PURCHASE (CCode, BCode, PurchaseDate, Review)
```

Assume the following cardinalities:

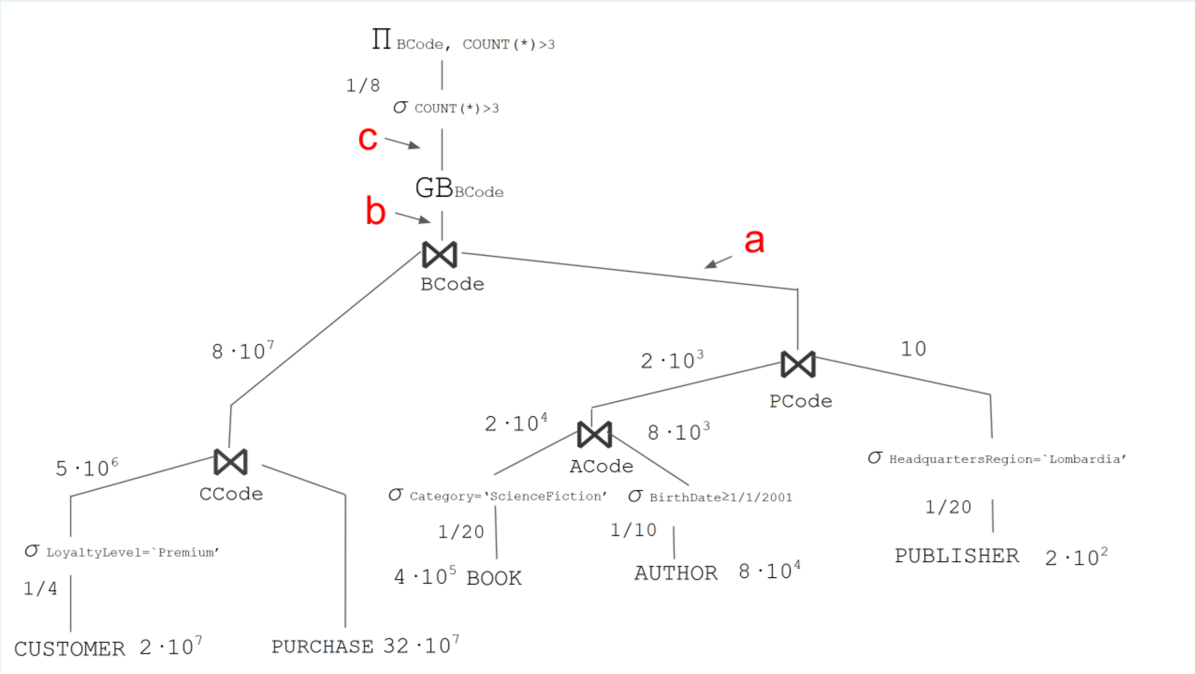
- card(BOOK) = $4 \cdot 10^5$ tuples,
  Distinct values of Category = 20
  Reduction factor of Price > 20€ = 1/15
- card(AUTHOR) = $8 \cdot 10^4$ tuples,
  Distinct values of Country = 80
  MIN(BirthDate) = 1/1/1920, MAX(BirthDate) = 31/12/2009
- card(PUBLISHER) = $2 \cdot 10^2$ tuples,
  Distinct values of HeadquartersRegion = 20
- card(CUSTOMER) = $2 \cdot 10^7$ tuples,
  MIN(BirthDate) = 1/1/1945, MAX(BirthDate) = 31/12/2014
  Distinct values of ResidenceProvince = 100
  Distinct values of LoyaltyLevel = 4
- card(PURCHASE)= $32 \cdot 10^7$ tuples,

Furthermore, assume the following reduction factor for the having clauses:
Having COUNT(*)>3 = 1/8

Consider the following query:

```
SELECT B.BCode, COUNT(*)
FROM   CUSTOMER C, PURCHASE P, BOOK B, AUTHOR A, PUBLISHER PU
WHERE  P.CCode = C.CCode
       AND B.ACode = A.ACode
       AND B.PCode = PU.PCode
       AND P.BCode = B.BCode
       AND C.LoyaltyLevel = `Premium'
       AND PU.ResidenceProvince = `Lombardia'
       AND A.BirthDate >= 1/1/2001
       AND B.Category = `ScienceFiction'
GROUP BY B.BCode
HAVING COUNT(*) > 3
```

The figure below represents the query tree for the query above.



Specify the cardinality of each node indicated by the red letters (a,b,c) in the figure. There is one right answer for each node a,b,c.

- [ ] a. b: $5 \cdot 10^5$
- [ ] b. c: $5 \cdot 10^3$
- [ ] c. b: $4 \cdot 10^5$
- [ ] d. c: $2 \cdot 10^3$
- [x] e. b: $2 \cdot 10^4$ ✔
- [ ] f. a: $5 \cdot 10^3$
- [ ] g. b: $5 \cdot 10^4$
- [ ] h. c: $5 \cdot 10^4$
- [x] i. c: 10^2 ✔
- [ ] j. a: $2 \cdot 10^3$
- [x] k. a: 10^2 ✔
- [ ] l. a: $5 \cdot 10^2$

Risposta corretta.

Le risposte corrette sono:
a: 10^2,

b: $2 \cdot 10^4$,

c: 10^2

Domanda 4
Risposta corretta
Punteggio ottenuto 1,00 su 1,00

Indexes (1 point, -15% penalty for each wrong answer)

The following tables are provided:

```
BOOK (BCode, Title, ACode, PCode, Category, PublicationYear, Price)
AUTHOR (ACode, FirstName, LastName, Country, BirthDate)
PUBLISHER (PCode, PName, FoundationDate, HeadquartersRegion)
CUSTOMER (CCode, FirstName, LastName, BirthDate, ResidenceProvince, LoyaltyLevel)
PURCHASE (CCode, BCode, PurchaseDate, Review)
```

Assume the following cardinalities:

- card(BOOK) = $4 \cdot 10^5$ tuples,
  Distinct values of Category = 20
  Reduction factor of Price > 20€ = 1/15
- card(AUTHOR) = $8 \cdot 10^4$ tuples,
  Distinct values of Country = 80
  MIN(BirthDate) = 1/1/1920, MAX(BirthDate) = 31/12/2009
- card(PUBLISHER) = $2 \cdot 10^2$ tuples,
  Distinct values of HeadquartersRegion = 20
- card(CUSTOMER) = $2 \cdot 10^7$ tuples,
  MIN(BirthDate) = 1/1/1945, MAX(BirthDate) = 31/12/2014
  Distinct values of ResidenceProvince = 100
  Distinct values of LoyaltyLevel = 4
- card(PURCHASE)= $32 \cdot 10^7$ tuples,

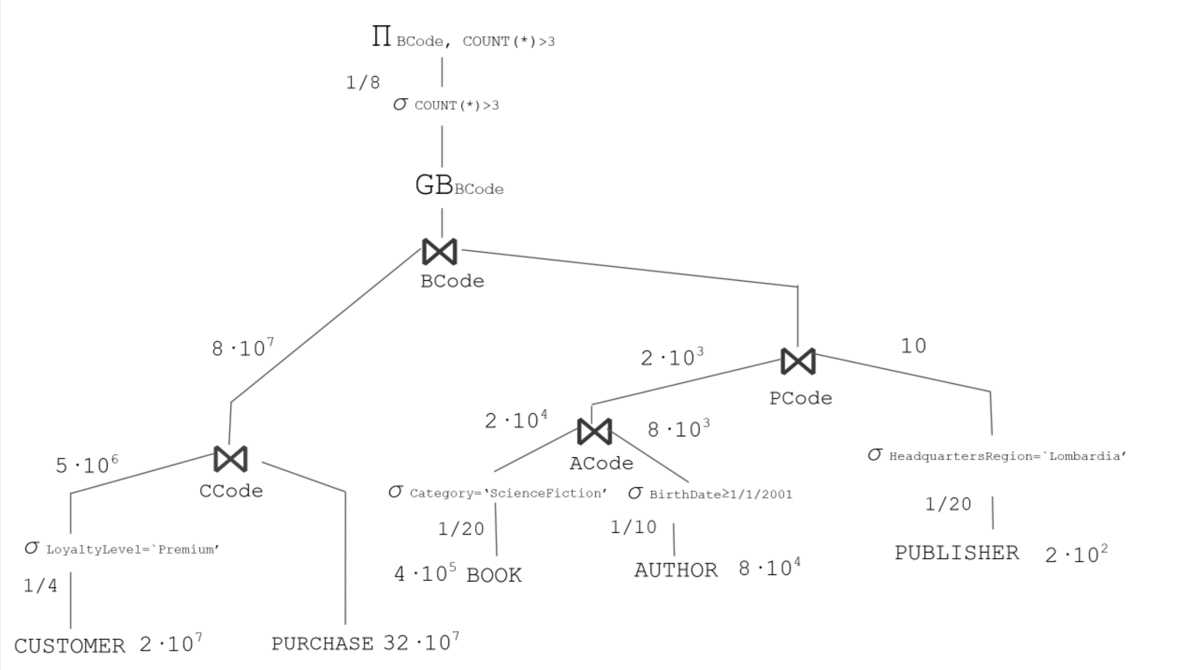
Furthermore, assume the following reduction factor for the having clauses:

Having COUNT(*)>3 = 1/8

Consider the following query:

```
SELECT B.BCode, COUNT(*)
FROM   CUSTOMER C, PURCHASE P, BOOK B, AUTHOR A, PUBLISHER PU
WHERE  P.CCode = C.CCode
       AND B.ACode = A.ACode
       AND B.PCode = PU.PCode
       AND P.BCode = B.BCode
       AND C.LoyaltyLevel = `Premium’
       AND PU.ResidenceProvince = `Lombardia’
       AND A.BirthDate >= 1/1/2001
       AND B.Category = `ScienceFiction’
GROUP BY B.BCode
HAVING COUNT(*) > 3
```

The figure below represents the query tree for the query above.



Select one or more secondary physical structures to increase query performance (if possible) among the options below. You can select multiple correct answers.

- a. CREATE INDEX IndexB ON PURCHASE(PurchaseDate) - B+-Tree
- b. CREATE INDEX IndexD ON CUSTOMER(LoyaltyLevel) - B+-Tree
- c. CREATE INDEX IndexI ON PUBLISHER(HeadquartersRegion) - HASH
- d. CREATE INDEX IndexF ON BOOK(Category) - B+-Tree
- e. None – the additional physical structures would not improve the query performance
- f. CREATE INDEX IndexC ON CUSTOMER(LoyaltyLevel) - HASH
- g. CREATE INDEX IndexA ON PURCHASE(PurchaseDate) - HASH
- h. CREATE INDEX IndexJ ON PUBLISHER(HeadquartersRegion) - B+-Tree
- i. CREATE INDEX IndexH ON AUTHOR(BirthDate) - B+-Tree ✔
- j. CREATE INDEX IndexG ON AUTHOR(BirthDate) - HASH
- k. CREATE INDEX IndexE ON BOOK(Category) - HASH ✔

Risposta corretta.

Le risposte corrette sono: CREATE INDEX IndexE ON BOOK(Category) - HASH, CREATE INDEX IndexH ON AUTHOR(BirthDate) - B+-Tree

Domanda 5
Risposta corretta
Punteggio ottenuto 1,80 su 1,80

Measures (1.8 points, -15% penalty for each wrong answer)

We want to analyze the information relating to failure management for traffic lights located in urban areas in italian cities.

The location of the traffic light where the failure occurred is known, including the neighborhood, district, city, province, and region, as well as whether it was located at a high-traffic intersection. The list of facilities available in each neighborhood such as schools, hospitals, children's play areas, etc, is also provided. Also the time since traffic light was installed is known (one value between less than 1 year, between 1 and 5 years, between 6 and 10 years, more than 10 years).

Failures are characterized by their causes (one or more values between 'Electrical', 'Environmental', 'Mechanical', 'External', 'Operational') and the indication of their severity level (one value between 'Low', 'Medium' and 'High'). Also the required type of intervention is available (one or more values between

‘Repair’, ‘Replacement’, ‘Software Update’). Companies involved in the repair of the failures are characterized by their geographical location (expressed in terms of city, region and country) and their size (one value between small, medium, large).

You want to store the time the failure report was received, expressed in terms of date, day of the week, month, month of the year, quarter (3M), four-month period (4M), six-month period (6M), and year. Also the time slot of the failure report is known (one value between 'morning', 'afternoon', 'evening', 'night’).

We want to analyze the average repair cost per traffic light and average repair duration per traffic light based on the following information:

- Neighborhood, district, city, province, and region where a traffic light failure occurred. The list of facilities (e.g., schools, hospitals, children’s play areas, etc.) available in each neighborhood is also provided.
- Indication of whether the failure occurred in a traffic light located at a high-traffic intersection (one value between ‘yes’ and ‘no’)
- Time since traffic light installation (one value between less than 1 year, between 1 and 5 years, between 6 and 10 years, more than 10 years)
- Indication of failure causes (one or more values between ‘Electrical’, ‘Environmental’, ‘Mechanical’, ‘External’, ‘Operational’)
- Indication of the failure severity level (one value between ‘Low’, ‘Medium’ and ‘High’)
- Indication of the required type of intervention (one or more value between ‘Repair’, ‘Replacement’, ‘Software Update’)
- Repair company, with indication of the geographical location expressed in terms of city, region and country, is known. The size of the repair company is also known (one value between small, medium, large)
- Date, month, quarter (3M), four-month period (4M), six-month period (6M) and year in which the failure report was received. For the date, the indication of the day of week is also known, and for the month the indication of the month of the year is available. Finally, the time slot of the failure report is known (one value between 'morning', 'afternoon', 'evening', 'night’).

Select, from the dimensions proposed below, those that meet the requirements described in the problem specifications.

- a. Minimum repair cost, Minimum duration of repair work, Total number of traffic lights
- b. Average repair cost per traffic light, Total duration of repair work, Total number of traffic lights
- c. Average repair cost per traffic light, Average repair duration per traffic light
- d. Total number of failure types, Total number of traffic lights, Total repair cost
- e. None of the other answers is correct
- f. Average repair cost per traffic light, Average repair duration per traffic light, Total number of traffic lights
- g. Total repair cost, Total duration of repair work, Total number of repair companies
- h. Average repair duration per traffic light, Total repair cost, Total number of traffic lights
- i. Total repair cost, Total duration of repair work, Total number of traffic lights ✔

Risposta corretta.

La risposta corretta è:
Total repair cost, Total duration of repair work, Total number of traffic lights

Domanda 6

Risposta non data

This question is not a part of the exam

You can use the text area below to write any note or draft (e.g. intermediate steps of an exercise).

Non valutata

Any text written below will not be considered toward the correction of the exam.

Domanda 7

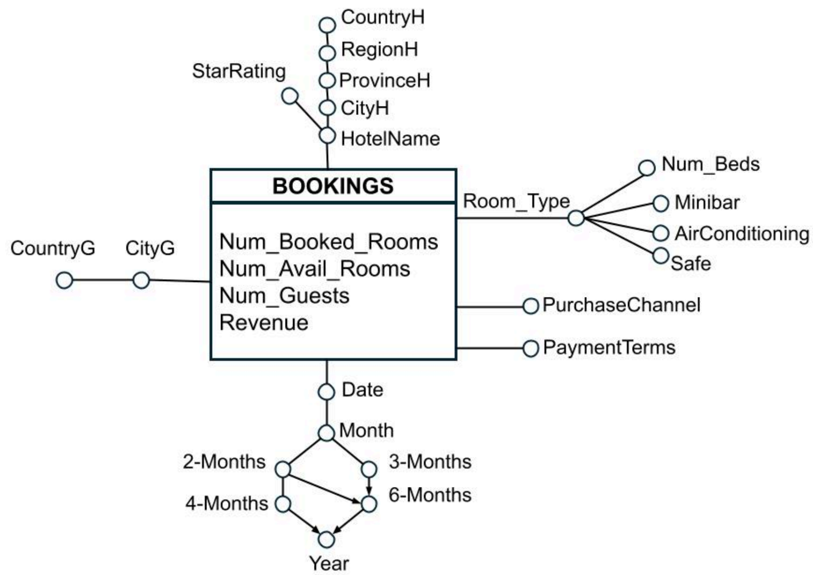
Completo

Punteggio ottenuto 4,00 su 4,00

Extended SQL (4 points)

The following data warehouse stores information related to hotel booking management on an online booking platform, analysing, every day, the number of rooms booked, the number of vacant rooms, the total number of guests and the total revenue. Hotels are characterised by a category described by the number of stars and a geographical hierarchy, while the type of room is described by the number of beds and the presence of amenities such as minibars, air conditioning and safes (configuration attribute). The city and state of origin of guests, payment terms (advance/at the property) and purchase channel (private/agency) are also taken into account.

The data warehouse is characterized by the following conceptual scheme and the corresponding logical scheme.



HOTEL (<u>HotelID</u>, HotelName, StarRating, CityH, ProvinceH, RegionH, CountryH)

ROOM-TYPE (<u>RTID</u>, Number_Beds, Minibar, AirConditioning, Safe)

TIME (<u>TimeID</u>, Date, Month, 2-Months, 3-Months, 4-Months, 6-Months, Year)

JUNK-PURCHASE (<u>IDJP</u>, PurchaseChannel, PaymentTerms)

GUEST_GEO (<u>GeoID</u>, CityG, CountryG)

BOOKINGS (<u>HotelID</u>, <u>RTID</u>, <u>TimeID</u>, <u>IDJP</u>, <u>GeoID</u>, Num_Booked_Rooms, Num_Avail_Rooms, Num_Guests, Revenue)

For each hotel province and for each 2-month period, compute:

- the percentage between the total number of booked rooms and the total number of rooms (considering both booked and available rooms)
- the percentage of revenue with respect to total revenue considering all hotels located in the same region;
- the cumulative revenue over successive 2-month periods, computed separately by year and by the region in which the hotel is located.

The analysis must be performed separately for each purchase channel.

SELECT ProvinceH, 2-Months, PurchaseChannel,

100*SUM(Num_Booked_Rooms)/(SUM(Num_Booked_Rooms) + SUM(Num_Avail_Rooms)),

100*SUM(Revenue)/SUM(SUM(Revenue)) OVER (PARTITION BY 2-Months, PurchaseChannel, RegionH),

SUM(SUM(Revenue)) OVER (PARTITION BY Year, RegionH, PurchaseChannel ORDER BY 2-Months ROWS UNBOUNDED PRECEDING)

FROM HOTEL H, TIME T, JUNK-PURCHASE JP, BOOKINGS B

WHERE B.HotelID = H.HotelID AND B.TimeID = T.TimeID AND B.IDJP = JP.IDJP

GROUP BY ProvinceH, 2-Months, PurchaseChannel, RegionH, Year

SELECT RegionH, 2-Month, PurchaseChannel, Year, 100 * SUM(Num_Booked_Rooms)/SUM(Num_Booked_Rooms+Num_Avail_Rooms)

100* SUM(Revenue) / SUM (SUM(Revenue)) OVER (PARTITION BY 2-Months, PurchaseChannel, RegionH)

SUM (SUM(Revenue)) OVER (PARTITION BY RegionH, Year, PurchaseChannel ORDER BY 2-Months ROWS UNBOUNDED PRECEDING)

FROM BOOKINGS B, HOTEL H, TIME T, JUNK-PURCHASE JP

WHERE <join conditions>

GROUP BY 2-Month, PurchaseChannel, RegionH, Year, ProvinceH

Commento:

Domanda 8

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

1 point (-15% for incorrect answer)

Documentation

- $match - Filters documents to include only those that match the specified conditions
  Syntax: $match: { <query> }
- $group -Groups documents according to a specified expression and applies accumulator expressions
  Syntax: $group: { _id: <expression>, <field>: { <accumulator>: <expression> } } }
- $unwind - Deconstructs an array field to output a document for each element.
  Syntax: $unwind: "$<field_name_array>"
- $sum - Computes the sum of numeric values (used within $group)
  Syntax: $sum: <expression>

Consider a collection called “products” that stores documents related to the shopping cart of an e-commerce website. Each document includes an array of products . An example of a document is shown below:

```
{
  _id: ObjectId("..."),
  id_order: "ORD-2024-001",
  id_customer: "CLI-12345",
 order_date: ISODate("2024-01-15T10:30:00Z"),
  status: "completed",
  products: [
    {
      id_product: "PROD-001",
      product_name: "Laptop",
      category: "Electronic",
      quantity: 1,
      price: 1200
    },
    {
      id_product: "PROD-045",
      product_name: "Mouse",
      category: "Electronic",
      quantity: 2,
      price: 25
    }
  ],
 total_price: 1250,
  shipping_country: "Italy"
}
```

Which query can be used to calculate the total number of products sold for each category, considering only orders with status “completed”, and showing as output the category and the total number of products sold?

a. None of the other answers is correct

b.
```
db.orders.aggregate([
  { $match: { status: "completed" } },
  { $unwind: "$products" },
  { $group: { _id: "$products.category", total_products: { $sum: "$products.quantity" } } }
])
```

c.
```
db.orders.aggregate([
  { $match: { status: "completed" } },
  { $unwind: "$products.category" },
  { $group: { _id: "$products.category", total_products: { $sum: "$products.quantity" } } }
])
```

d.
```
db.orders.aggregate([
  { $unwind: "$products" },
  { $group: { id: "$products.category", total_products: { $sum: "$products.quantity" }, } },
  { $match: { status: "completed" } },
])
```

e.
```
db.orders.aggregate([
  { $match: { status: "completed" } },
  { $group: { _id: "$products.category", total_products: { $sum: "$products.quantity" }, } },
  { $unwind: "$products" },
])
```

f.
```
db.orders.aggregate([
  { $unwind: "$products.category" },
  { $group: { _id: "$products.category", total_products: { $sum: "$products.quantity" }, } },
  { $match: { status: "completed" } },
])
```

Risposta corretta.

La risposta corretta è:

```
db.orders.aggregate([
  { $match: { status: "completed" } },
  { $unwind: "$products" },
  { $group: { _id: "$products.category", total_products: { $sum: "$products.quantity" } } }
])
```

Domanda 9
Risposta corretta
Punteggio ottenuto 1,50 su 1,50

1.5 points (-15% penalty for each incorrect answer)

We consider a classification problem consisting of four classes: $\{A, B, C, D\}$.

The following labeled training points are given in the two dimensional space:

- Class A: $((-1,1),(-2,2),(-5,5))$
- Class B: $((-1,-1),(-2,-2),(-5,-5))$
- Class C: $((1,-1),(2,-2),(5,-5))$
- Class D: $((0,0),(2,2),(5,5))$

A new point $p=(1,1)$ must be classified using the k-NN with $k=4$ and the Manhattan distance.

The Manhattan distance between two points x and y is defined as $dist(x,y)=\sum_i |x_i - y_i|$.

What is the class assigned to the new point p?

a. Class A

b. Class D ✔

c. Class C

d. Class B

e. There is not enough information about the algorithm’s hyperparameters to determine the answer.

f. The classification is ambiguous.

Risposta corretta.

La risposta corretta è:
Class D

Domanda 10
Risposta corretta
Punteggio ottenuto 1,00 su 1,00

Group by anticipation (1 point, -15% penalty for each wrong answer)

The following tables are provided:

```
BOOK (BCode, Title, ACode, PCode, Category, PublicationYear, Price)
AUTHOR (ACode, FirstName, LastName, Country, BirthDate)
PUBLISHER (PCode, PName, FoundationDate, HeadquartersRegion)
CUSTOMER (CCode, FirstName, LastName, BirthDate, ResidenceProvince, LoyaltyLevel)
PURCHASE (CCode, BCode, PurchaseDate, Review)
```

Assume the following cardinalities:

- card(BOOK) = $4{\cdot}10^5$ tuples,
  Distinct values of Category = 20
  Reduction factor of Price > 20€ = 1/15
- card(AUTHOR) = $8{\cdot}10^4$ tuples,
  Distinct values of Country = 80
  MIN(BirthDate) = 1/1/1920, MAX(BirthDate) = 31/12/2009
- card(PUBLISHER) = $2{\cdot}10^2$ tuples,
  Distinct values of HeadquartersRegion = 20
- card(CUSTOMER) = $2{\cdot}10^7$ tuples,
  MIN(BirthDate) = 1/1/1945, MAX(BirthDate) = 31/12/2014

Distinct values of ResidenceProvince = 100
Distinct values of LoyaltyLevel = 4

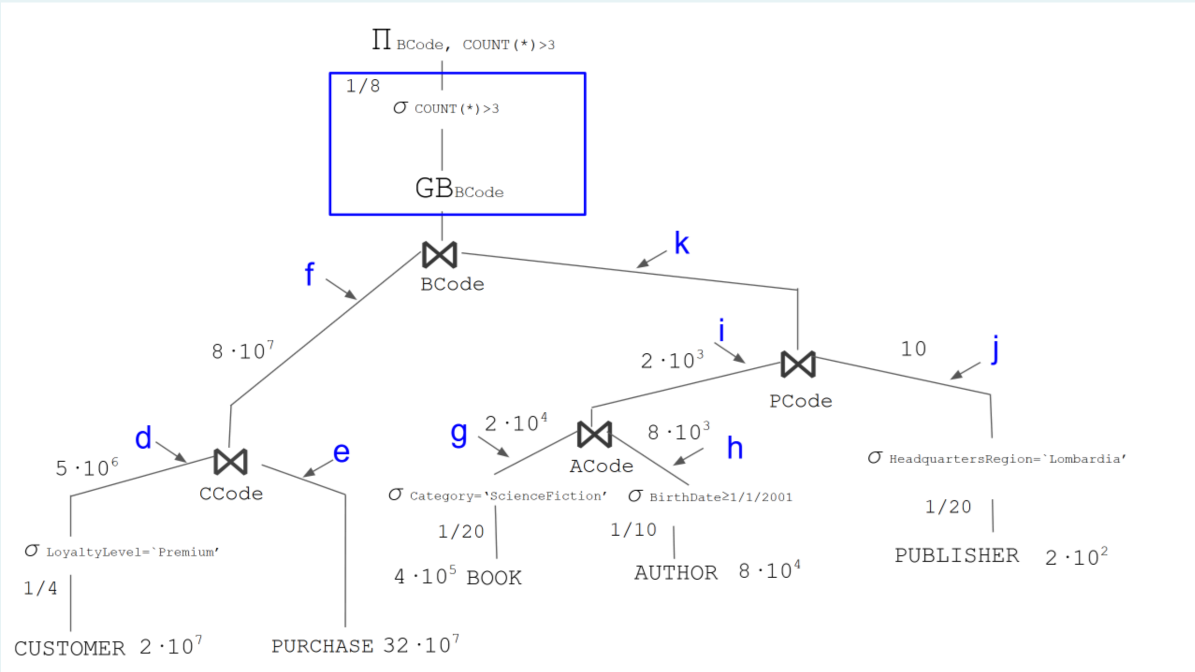
- card(PURCHASE)= $32 \cdot 10^7$ tuples,

Furthermore, assume the following reduction factor for the having clauses:
Having COUNT(*)>3 = 1/8

Consider the following query:

```
SELECT B.BCode, COUNT(*)
FROM   CUSTOMER C, PURCHASE P, BOOK B, AUTHOR A, PUBLISHER PU
WHERE  P.CCode = C.CCode
       AND B.ACode = A.ACode
       AND B.PCode = PU.PCode
       AND P.BCode = B.BCode
       AND C.LoyaltyLevel = `Premium'
       AND PU.ResidenceProvince = `Lombardia'
       AND A.BirthDate >= 1/1/2001
       AND B.Category = `ScienceFiction'
GROUP BY B.BCode
HAVING COUNT(*) > 3
```

The figure below represents the query tree for the query above.



Analyze the anticipation of the GROUP BY group by B.BCode having COUNT(*)>3 represented in the box. Select the solution that allows for the maximum efficiency in executing the query (if it exists)

- a. It is possible to anticipate it in branch j
- b. It is possible to anticipate it in branch d
- c. It is possible to anticipate it in branch g
- d. It is possible to anticipate it in branch f ✔
- e. It is possible to anticipate it in branch i
- f. It is possible to anticipate it in branch h
- g. It is not possible to anticipate the Group BY group by B.BCode having COUNT(*)>3
- h. It is possible to anticipate it in branch k
- i. It is possible to anticipate it in branch e

Risposta corretta.

La risposta corretta è: It is possible to anticipate it in branch f

Domanda **11**

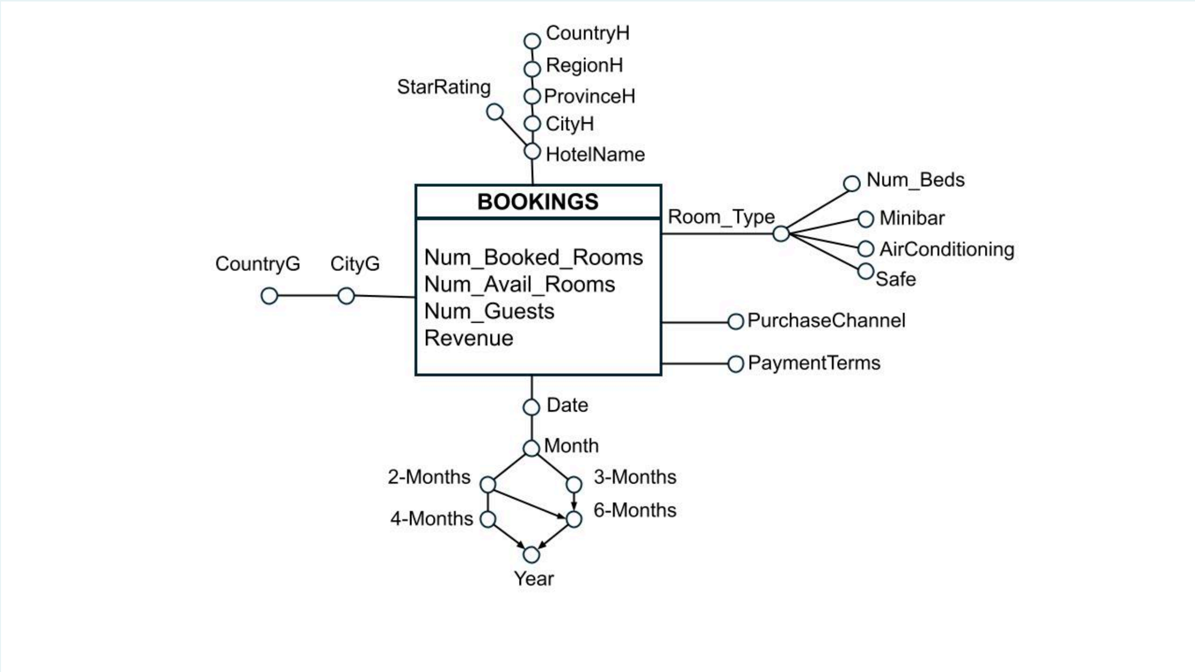
Completo

Punteggio ottenuto 5,00 su 5,00

Materialized view (5 points)

The following data warehouse stores information related to hotel booking management on an online booking platform, analysing, every day, the number of rooms booked, the number of vacant rooms, the total number of guests and the total revenue. Hotels are characterised by a category described by the number of stars and a geographical hierarchy, while the type of room is described by the number of beds and the presence of amenities such as minibars, air conditioning and safes (configuration attribute). The city and state of origin of guests, payment terms (advance/at the property) and purchase channel (private/agency) are also taken into account.

The data warehouse is characterized by the following conceptual scheme and the corresponding logical scheme.



HOTEL (<u>HotelID</u>, HotelName, StarRating, CityH, ProvinceH, RegionH, CountryH)

ROOM-TYPE (<u>RTID</u>, Number_Beds, Minibar, AirConditioning, Safe)

TIME (<u>TimeID</u>, Date, Month, 2-Months, 3-Months, 4-Months, 6-Months, Year)

JUNK-PURCHASE (<u>IDJP</u>, PurchaseChannel, PaymentTerms)

GUEST_GEO (<u>GeoID</u>, CityG, CountryG)

BOOKINGS (<u>HotelID</u>, <u>RTID</u>, <u>TimeID</u>, <u>IDJP</u>, <u>GeoID</u>, Num_Booked_Rooms, Num_Avail_Rooms, Num_Guests, Revenue)

Given the above logical scheme, consider the following queries of interest:

a. Considering the rooms that offer both minibar service (attribute MiniBar) and a safe (attribute Safe), display the total revenue, separately by hotel region (attribute RegionH) and by 2-Month period. Additionally, display the ranking based on total revenue, separately for each 2-Month period
b. For hotels in the city of Turin, show the cumulative value of revenue over two months (attribute 2-Months), separately for each year.
c. Separately by 3-Month period and by hotel region, for rooms equipped with a safe, display the percentage of revenue with respect to the total revenue by hotel state and year.

Given the above logical schema, answer the following requests:

1. Define a materialized view with the CREATE MATERIALIZED VIEW, that can be used to efficiently answer all three of the above queries (i.e., a, b, c). Specifically, define the query in SQL associated with BLOCK A in the following instruction

```
CREATE MATERIALIZED VIEW ViewBookings
BUILD IMMEDIATE
REFRESH FAST ON COMMIT
AS
Block A
```

---

2. Define the **minimal set** of attributes that allows identifying the tuples belonging to the materialized view ViewBookings.
3. Assume that the management of the materialized view (derived table) is carried out by means of triggers. Write the trigger to propagate to the ViewBookings materialized view the changes due to the insertion of a new record into the BOOKINGS fact table

1. Block A:

```
SELECT CityH, RegionH, StateH, Minibar, Safe, 2-Months, 3-Months, Year, SUM(Revenue) AS TotR
FROM HOTEL H, ROOM-TYPE RT, TIME T, BOOKINGS B
WHERE B.HotelID = H.HotelID AND B.RTID = RT.RTID AND B.TimeID = T.TimeID
GROUP BY CityH, RegionH, StateH, Minibar, Safe, 2-Months, 3-Months, Year;
```

2. Minimal set:

CityH, Minibar, Safe, 2-Months, 3-Months.

3. Trigger:

```
CREATE OR REPLACE TRIGGER TriggerViewBookings
AFTER INSERT ON BOOKINGS
FOR EACH ROW
DECLARE
varCityH, varRegionH, varStateH VARCHAR(10);
varMinibar, varSafe BOOLEAN;
var2M, var3M, varY DATE;
N INTEGER;
BEGIN
SELECT CityH, RegionH, StateH INTO varCityH, varRegionH, varStateH
FROM HOTEL
WHERE HotelID = :NEW.HotelID;
SELECT Minibar, Safe INTO varMinibar, varSafe
FROM ROOM-TYPE
WHERE RTID = :NEW.RTID;
SELECT 2-Months, 3-Months, Year INTO var2M, var3M, varY
FROM TIME
WHERE TimeID = :NEW.TimeID;
SELECT COUNT(*) INTO N
FROM ViewBookings
WHERE CityH = varCityH AND Minibar = varMinibar AND Safe = varSafe AND 2-Months = var2M AND 3-Months = var3M;
IF (N > 0) THEN
```

```
UPDATE ViewBookings
SET TotR = TotR + :NEW.Revenue
WHERE CityH = varCityH AND Minibar = varMinibar AND Safe = varSafe AND 2-Months = var2M AND 3-Months = var3M;
ELSE
INSERT INTO ViewBookings(CityH, RegionH, StateH, Minibar, Safe, 2-Months, 3-Months, Year, TotR)
VALUES(varCityH, varRegionH, varStateH, varMinibar, varSafe, var2M, var3M, varY, :NEW.Revenue);
END IF;
END;
```

Query of interest:

```
(a) SELECT RegionH, 2-Mesi, SUM(Revenue), RANK() OVER (PARTITION BY 2-Months ORDER BY SUM(Revenue))
FROM HOTEL H, TIME T, ROOM_TYPE RT, BOOKINGS B
WHERE <join conditions> AND MiniBar = ‘Y’ AND Safe = ‘Y’
GROUP BY RegionH, 2-Months;

(b) SELECT 2-Months, Year, SUM(SUM(Revenue)) OVER (PARTITION BY Year ORDER BY 2-Months
                                          ROWS UNBOUNDED PRECEDING)
FROM HOTEL H, TIME T, BOOKINGS B
WHERE <join conditions> AND CityH = ‘Torino’
GROUP BY 2-Months, Year;

(c)
SELECT 3-Months, RegionH, 100*SUM(Revenue)/SUM(SUM((Revenue))
OVER (PARTITION BY CountryH, Year)
FROM HOTEL H, TIME T, BOOKINGS B
WHERE <join conditions> AND Safe = ‘Y’
GROUP BY 3-Months, Year, RegionH, CountryH;
```

1. Block A - Materialized view

```
SELECT 2-Months, 3-Months, Year, CityH, RegionH, CountryH, MiniBar, Safe, SUM(Revenue) AS TotRevenue
FROM HOTEL H, TIME T, ROOM_TYPE RT, BOOKINGS B
WHERE H.HotelID = B.HotelID AND RT.IDRT = B.IDRTAND T.TimeID = B.TimeID
GROUP BY 2-Months, 3-Months, Year, CityH, RegionH, CountryH, MiniBar, Safe
```

2. Identifier: 2-Months, 3-Months, CityH, MiniBar, Safe

3. Trigger

```
CREATE OR REPLACE TRIGGER ViewBOOKINGS
AFTER INSERT ON PRENOTAZIONI
FOR EACH ROW
DECLARE
Var2M, Var3M, VarYear DATE;
VarCityH, VarRegionH, VarCountryH, VARCHAR(10); VarMiniBar, VarSafe BOOLEAN;
N INTEGER;
BEGIN
```

```
        SELECT 2-Months, 3-Months, Year INTO Var2M, Var3M, VarYear
        FROM TIME
        WHERE TIMEID= :NEW.TIMEID;

        SELECT MiniBar, Safe INTO VarMiniBar, VarSafe
        FROM ROOM-TYPE
        WHERE RTID = :NEW.RTID;

        SELECT CityH, RegionH, CountryH INTO VarCityH, VarRegionH, VarCountryH
        FROM HOTEL
        WHERE HotelID= :NEW.HotelID;
SELECT COUNT(*) INTO N
FROM ViewBookings
WHERE 2-Months = Var2M AND 3-Months = Var3M AND CityH = VarCityH
AND MiniBar = VarMiniBar AND Safe= VarSafe;

IF N>0 THEN
        UPDATE ViewBookings
        SET TotRevenue = TotRevenue + :NEW.Revenue
         WHERE 2-Months = Var2M AND 3-Months = Var3M AND CityH = VarCityH AND MiniBar = VarMiniBar
AND
        Cassaforte = VarCassaforte;
ELSE
        INSERT INTO ViewBookings(...) VALUES (Var2M, Var3M, VarYear, VarCityH, VarRegionH, VarCountryH,
VarMiniBar, VaSafe, :NEW.Revenue);
END IF;
END;
```

Commento:

Domanda 12
Risposta corretta
Punteggio ottenuto 1,50 su 1,50

1.5 points, -15% penalty for each wrong answer

Notation

RN(x): read of object x by transaction Tn

WN(x): write of object x by transaction Tn

Exercise text

Consider the following schedule S of four transactions:

S=W0(x),R1(x),W1(y),R2(y),W2(x), R0(y)

Verify whether schedule S is conflict-serializable. If so, select the equivalent serial schedule

- a. S is not conflict-serializable ✔
- b. R2(y),W2(x),W0(x),R0(y),R1(x),W1(y)
- c. R1(x),W1(y),W0(x),R0(y),R2(y),W2(x)

- d. R1(x),W1(y),R2(y),W2(x),W0(x),R0(y)
- e. W0(x), R0(y), R1(x), W1(y), R2(y), W2(x)
- f. W0(x),R1(x),W1(y),R0(y),R2(y),W2(x)

Risposta corretta.

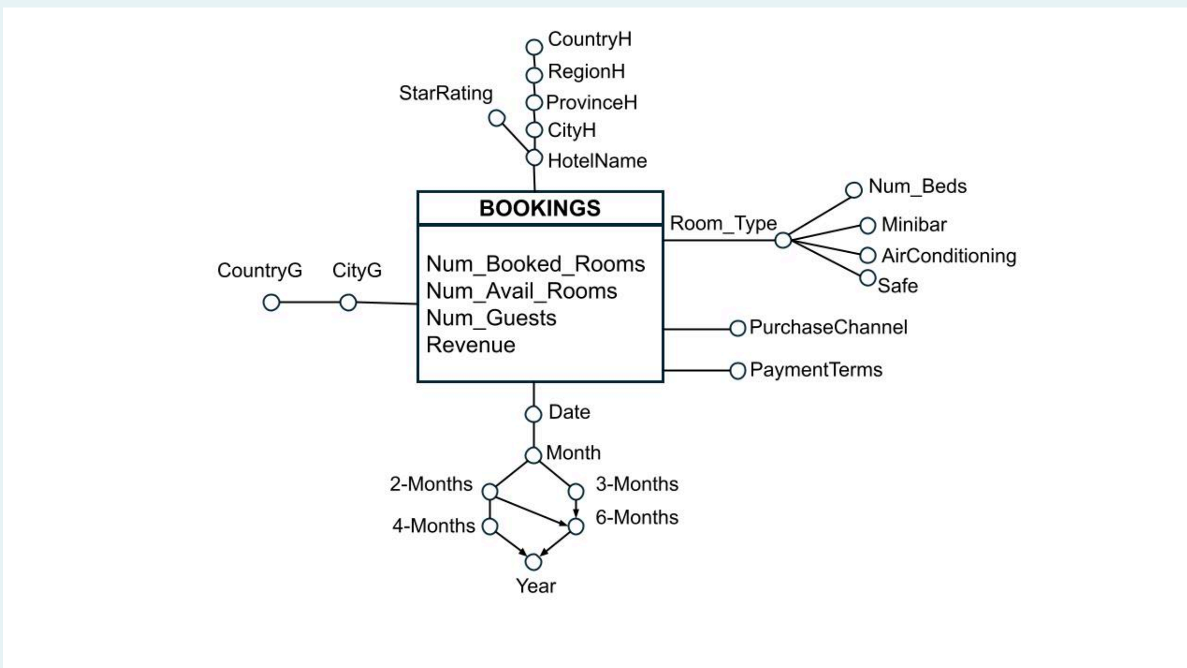
La risposta corretta è:
S is not conflict-serializable

Domanda 13
Completo
Punteggio ottenuto 3,00 su 3,00

Extended SQL (3 points)

The following data warehouse stores information related to hotel booking management on an online booking platform, analysing, every day, the number of rooms booked, the number of vacant rooms, the total number of guests and the total revenue. Hotels are characterised by a category described by the number of stars and a geographical hierarchy, while the type of room is described by the number of beds and the presence of amenities such as minibars, air conditioning and safes (configuration attribute). The city and state of origin of guests, payment terms (advance/at the property) and purchase channel (private/agency) are also taken into account.

The data warehouse is characterized by the following conceptual scheme and the corresponding logical scheme.



HOTEL (<u>HotelID</u>, HotelName, StarRating, CityH, ProvinceH, RegionH, CountryH)

ROOM-TYPE (<u>RTID</u>, Number_Beds, Minibar, AirConditioning, Safe)

TIME (<u>TimeID</u>, Date, Month, 2-Months, 3-Months, 4-Months, 6-Months, Year)

JUNK-PURCHASE (<u>IDJP</u>, PurchaseChannel, PaymentTerms)

GUEST_GEO (<u>GeoID</u>, CityG, CountryG)

BOOKINGS (<u>HotelID</u>, <u>RTID</u>, <u>TimeID</u>, <u>IDJP</u>, <u>GeoID</u>, Num_Booked_Rooms, Num_Avail_Rooms, Num_Guests, Revenue)

For each city of origin of the guest and for each 6-Month period, compute:

- the average number of guests per booked room;
- the percentage of revenue relative to the total revenue considering the guest’s country of origin;
- the total revenue, computed separately by year and by the guest’s country of origin.

Assign to each record a rank, computed separately for each year, based on total revenue

SELECT CityG, 6-Months, SUM(Num_Guests)/SUM(Num_Booked_Rooms),

100*SUM(Revenue)/SUM(SUM(Revenue)) OVER (PARTITION BY 6-Months, CountryG),

SUM(SUM(Revenue)) OVER (PARTITION BY Year, CountryG),

RANK () OVER (PARTITION BY Year ORDER BY SUM(Revenue))

FROM TIME T, GUEST_GEO G, BOOKINGS B

WHERE B.TimeID = T.TimeID AND B.GeoID = G.GeoID

GROUP BY CityG, 6-Months, CountryG, Year

SELECT CityH, 6-Months,

SUM(Num_Guests)/SUM(Num_Booked_Rooms)

100* SUM(Revenue) / SUM (SUM(Revenue)) OVER (PARTITION BY 6-Months, CountryG),

SUM (SUM(Revenue)) OVER (PARTITION BY Year, CountryG),

RANK() OVER(PARTITION BY Year ORDER BY SUM(Revenue))

FROM BOOKINGS B, GEO-GUESTS, TIME T

WHERE <join conditions>

GROUP BY CityG, 6-Months, Year, CountryG

Commento:

Domanda
14
Risposta corretta
Punteggio ottenuto 2,00 su 2,00

2 points (-15% penalty for each incorrect answer)

Consider the following transactional database:

TID	Transaction
1	{A, B, C, D, E}
2	{A, B, C, E, F}
3	{B, C, D, E}
4	{A, B, D, E, F, G}
5	{A, B, C, D, E, F, G}
6	{B, C, D, E, F, G}
7	{A, B, C, E, F}
8	{A, B, C, D, E, F, G}
9	{A, B, C, D, E}
10	{A, B, C, D, E, F, G}

Build the FP-Tree from the given set of transactions using a minimum support threshold MinSup ≥ 3. An itemset is frequent if it appears in at least 3 transactions.

In the case of items with equal support, lexicographic order should be used (for example, if items A and B have the same support, then A precedes B)

After constructing the FP-Tree, identify the patterns and their corresponding supports present in the G-CPB (G Conditional Pattern Base). Which of the following sets of pairs Pattern: Support represents the G-CPB? (where each pattern is a set of items).

- [ ] a. {B, E, C, D}: 1
- [x] b. {B, E, A, D, F}: 1 ✔
- [ ] c. {A, B, C, D, E, F}: 5
- [ ] d. {G, F, D, A, C, E, B}: 5
- [x] e. {B, E, C, A, D, F}: 3 ✔
- [ ] f. {B, E, C, A, D, F, G}: 3
- [ ] g. None of the other answers is correct
- [ ] h. {B, E, C, D, F}: 3
- [x] i. {B, E, C, D, F}: 1 ✔

Risposta corretta.

Le risposte corrette sono:
{B, E, A, D, F}: 1,

{B, E, C, A, D, F}: 3,

{B, E, C, D, F}: 1

Domanda
15
Risposta corretta

Conceptual schema (1.6 points, -15% penalty for each wrong answer)

Punteggio ottenuto 1,60 su 1,60

We want to analyze the information relating to failure management for traffic lights located in urban areas in italian cities.

The location of the traffic light where the failure occurred is known, including the neighborhood, district, city, province, and region, as well as whether it was located at a high-traffic intersection. The list of facilities available in each neighborhood such as schools, hospitals, children's play areas, etc, is also provided. Also the time since traffic light was installed is known (one value between less than 1 year, between 1 and 5 years, between 6 and 10 years, more than 10 years).

Failures are characterized by their causes (one or more values between 'Electrical', 'Environmental', 'Mechanical', 'External', 'Operational') and the indication of their severity level (one value between 'Low', 'Medium' and 'High'). Also the required type of intervention is available (one or more values between 'Repair', 'Replacement', 'Software Update'). Companies involved in the repair of the failures are characterized by their geographical location (expressed in terms of city, region and country) and their size (one value between small, medium, large).

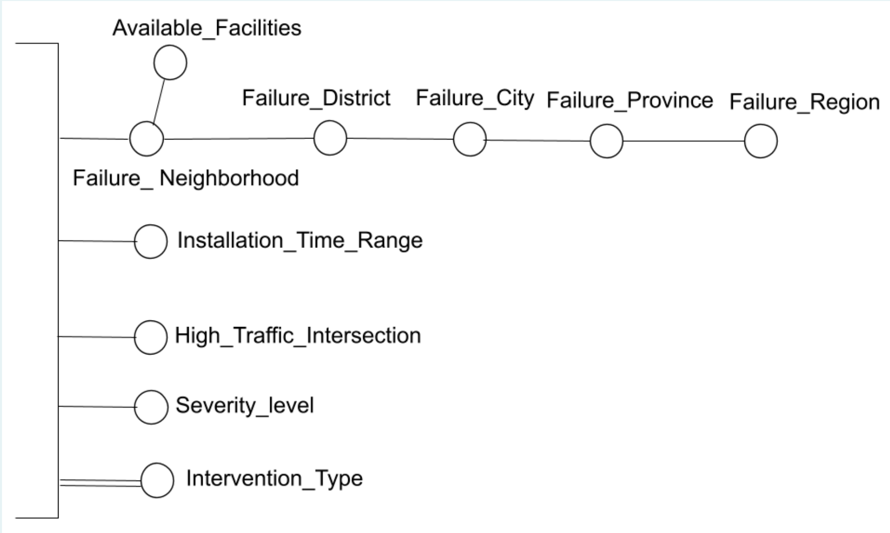
You want to store the time the failure report was received, expressed in terms of date, day of the week, month, month of the year, quarter (3M), four-month period (4M), six-month period (6M), and year. Also the time slot of the failure report is known (one value between 'morning', 'afternoon', 'evening', 'night').

We want to analyze the average repair cost per traffic light and average repair duration per traffic light based on the following information:

- Neighborhood, district, city, province, and region where a traffic light failure occurred. The list of facilities (e.g., schools, hospitals, children's play areas, etc.) available in each neighborhood is also provided.
- Indication of whether the failure occurred in a traffic light located at a high-traffic intersection (one value between 'yes' and 'no')
- Time since traffic light installation (one value between less than 1 year, between 1 and 5 years, between 6 and 10 years, more than 10 years)
- Indication of failure causes (one or more values between 'Electrical', 'Environmental', 'Mechanical', 'External', 'Operational')
- Indication of the failure severity level (one value between 'Low', 'Medium' and 'High')
- Indication of the required type of intervention (one or more value between 'Repair', 'Replacement', 'Software Update')
- Repair company, with indication of the geographical location expressed in terms of city, region and country, is known. The size of the repair company is also known (one value between small, medium, large)
- Date, month, quarter (3M), four-month period (4M), six-month period (6M) and year in which the failure report was received. For the date, the indication of the day of week is also known, and for the month the indication of the month of the year is available. Finally, the time slot of the failure report is known (one value between 'morning', 'afternoon', 'evening', 'night').

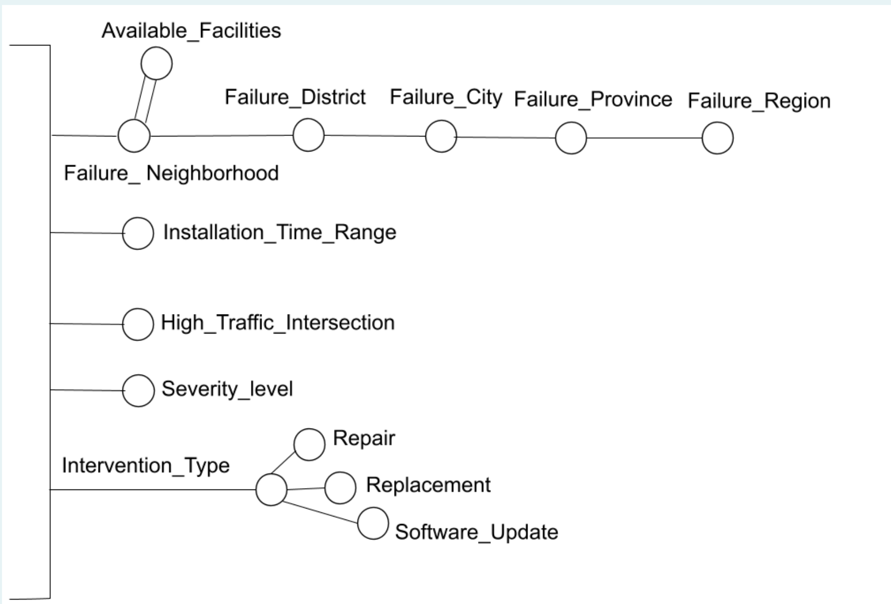
Select, from the dimensions proposed below, those that meet the requirements described in the problem specifications.

a.



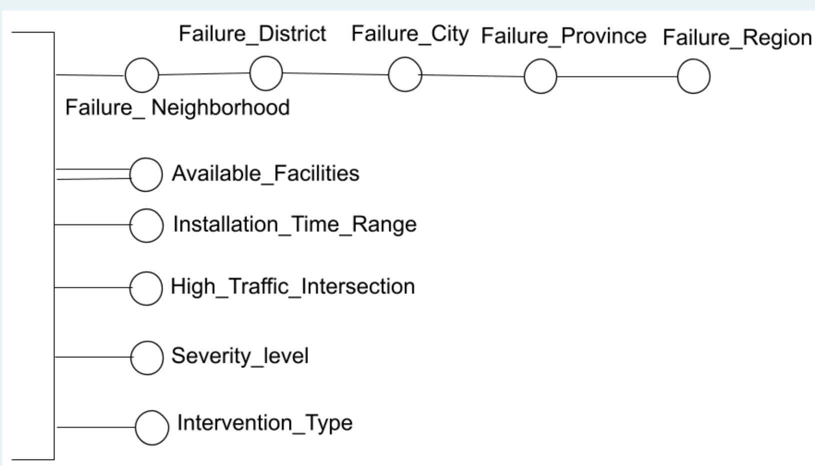
Available_Facilities
Failure_District
Failure_City
Failure_Province
Failure_Region
Failure_ Neighborhood
Installation_Time_Range
High_Traffic_Intersection
Severity_level
Intervention_Type

b.



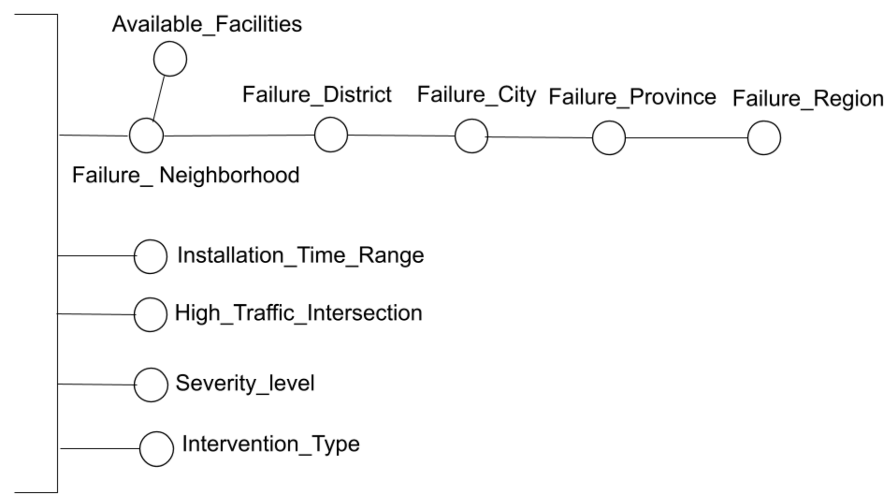
Available_Facilities
Failure_District
Failure_City
Failure_Province
Failure_Region
Failure_ Neighborhood
Installation_Time_Range
High_Traffic_Intersection
Severity_level
Repair
Intervention_Type
Replacement
Software_Update

c.

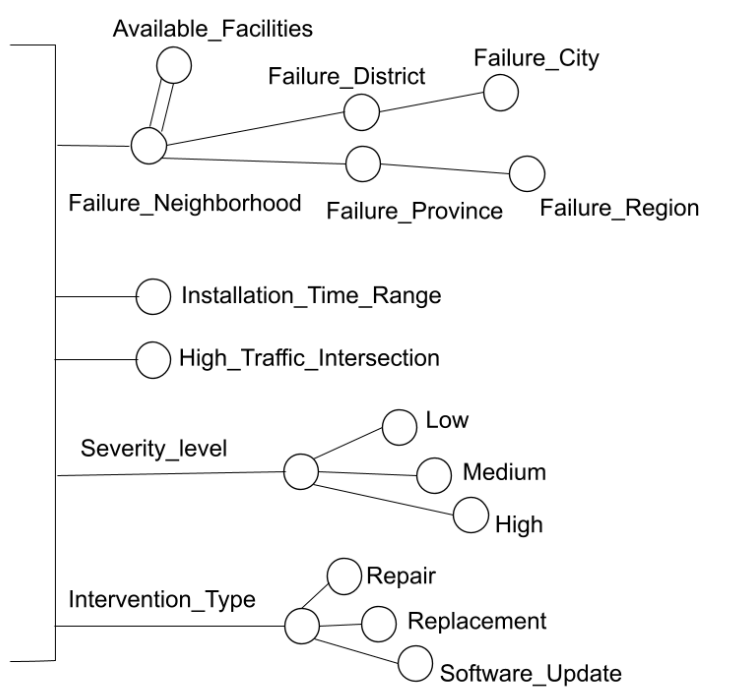


Failure_District
Failure_City
Failure_Province
Failure_Region
Failure_ Neighborhood
Available_Facilities
Installation_Time_Range
High_Traffic_Intersection
Severity_level
Intervention_Type

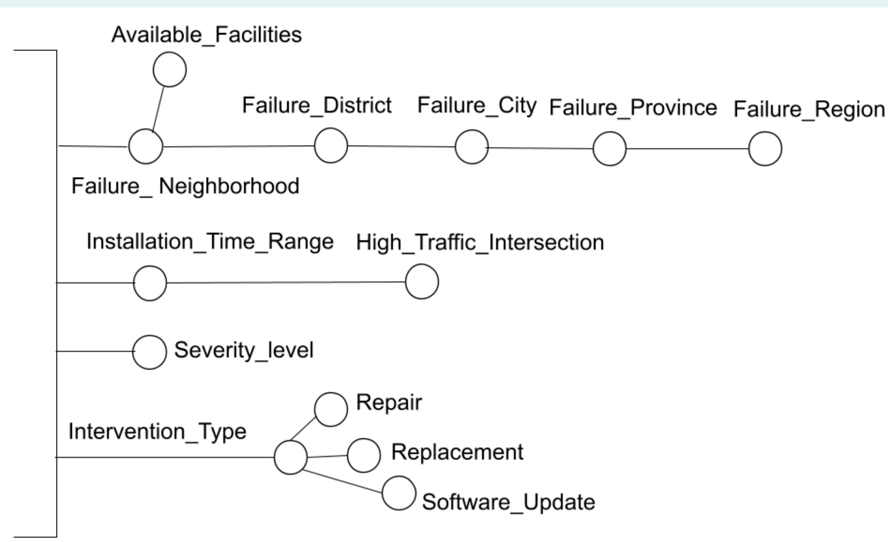
d.


Available_Facilities
Failure_District
Failure_City
Failure_Province
Failure_Region
Failure_ Neighborhood
Installation_Time_Range
High_Traffic_Intersection
Severity_level
Intervention_Type

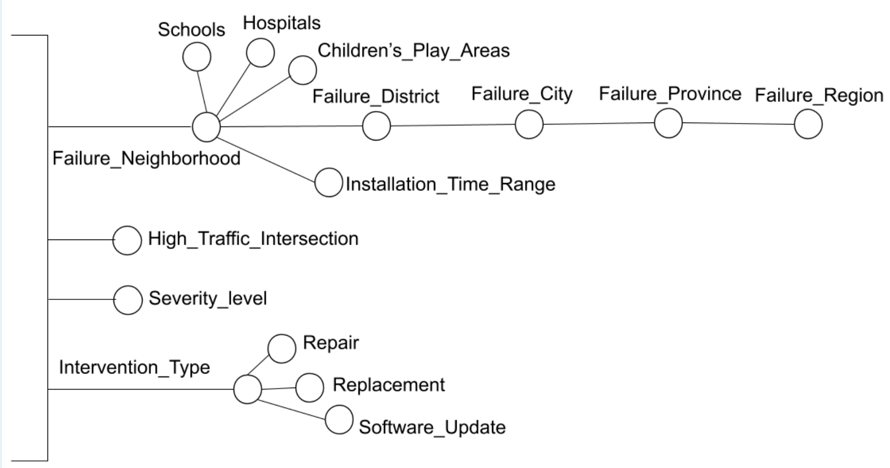
e.


Available_Facilities
Failure_District
Failure_City
Failure_Neighborhood
Failure_Province
Failure_Region
Installation_Time_Range
High_Traffic_Intersection
Severity_level
Low
Medium
High
Intervention_Type
Repair
Replacement
Software_Update

f.


Available_Facilities
Failure_District
Failure_City
Failure_Province
Failure_Region
Failure_ Neighborhood
Installation_Time_Range
High_Traffic_Intersection
Severity_level
Intervention_Type
Repair
Replacement
Software_Update

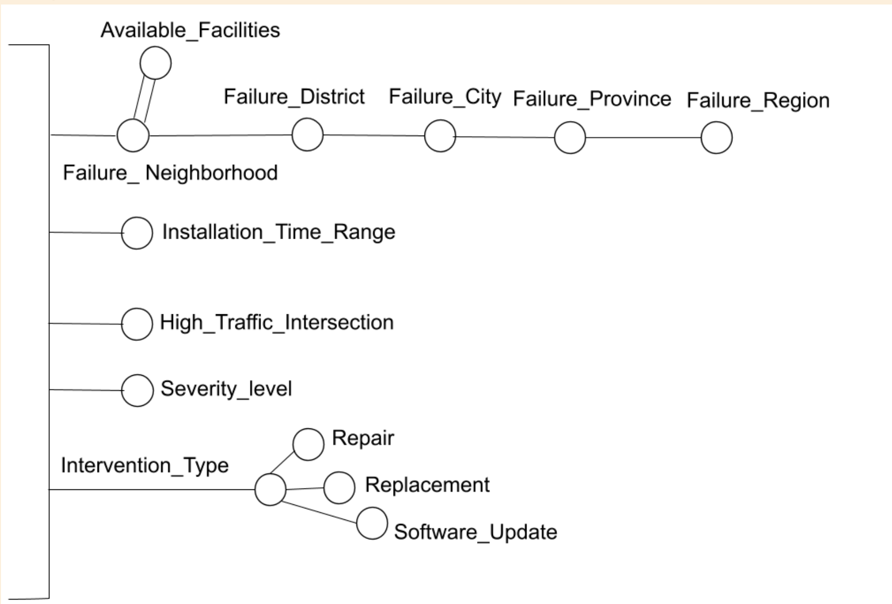
g.



h. None of the other answers is correct

Risposta corretta.

La risposta corretta è:



Domanda **16**

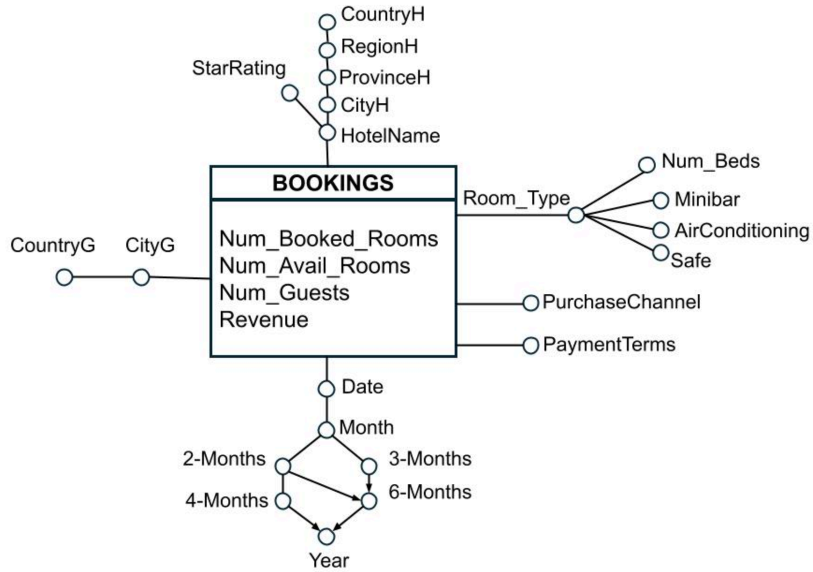
Completo

Punteggio ottenuto 3,75 su 4,00

Extended SQL (4 points)

The following data warehouse stores information related to hotel booking management on an online booking platform, analysing, every day, the number of rooms booked, the number of vacant rooms, the total number of guests and the total revenue. Hotels are characterised by a category described by the number of stars and a geographical hierarchy, while the type of room is described by the number of beds and the presence of amenities such as minibars, air conditioning and safes (configuration attribute). The city and state of origin of guests, payment terms (advance/at the property) and purchase channel (private/agency) are also taken into account.

The data warehouse is characterized by the following conceptual scheme and the corresponding logical scheme.



HOTEL (<u>HotelID</u>, HotelName, StarRating, CityH, ProvinceH, RegionH, CountryH)

ROOM-TYPE (<u>RTID</u>, Number_Beds, Minibar, AirConditioning, Safe)

TIME (<u>TimeID</u>, Date, Month, 2-Months, 3-Months, 4-Months, 6-Months, Year)

JUNK-PURCHASE (<u>IDJP</u>, PurchaseChannel, PaymentTerms)

GUEST_GEO (<u>GeoID</u>, CityG, CountryG)

BOOKINGS (<u>HotelID</u>, <u>RTID</u>, <u>TimeID</u>, <u>IDJP</u>, <u>GeoID</u>, Num_Booked_Rooms, Num_Avail_Rooms, Num_Guests, Revenue)

Considering room types equipped with air conditioning and a safe, for each 4-month period and for each hotel name, compute:

- the average monthly revenue;
- the ratio between revenue generated from bookings and total revenue considering the hotel's star rating;
- total revenue, computed separately by year and by the hotel's star rating.

Perform the analysis separately by payment terms and by the guest's country of origin.

SELECT 4-Months, HotelName, PaymentTerms, CountryG,

SUM(Revenue)/COUNT(DISTINCT Month),

(SUM(Revenue)/SUM(Num_Booked_Rooms))/SUM(SUM(Revenue)) OVER (PARTITION BY 4-Months, PaymentTerms, CountryG, StarRating),

SUM(SUM(Revenue)) OVER (PARTITION BY PaymentTerms, CountryG, Year, StarRating)

FROM HOTEL H, ROOM-TYPE RT, TIME T, JUNK-PURCHASE JP, GUEST_GEO G, BOOKINGS B

WHERE B.HotelID = H.HotelID AND B.RTID = RT.RTID AND B.TimeID = T.TimeID AND B.IDJP = JP.IDJP AND B.GeoID = G.GeoID

AND AirConditioning = 'Yes' AND Safe = 'Yes'

GROUP BY 4-Months, HotelName, PaymentsTerms, CountryG, StarRating, Year

SELECT 4-Months, HotelName, PaymentTerms, CountryG

SUM(Revenue) / COUNT(DISTINCT Month)

SUM(Revenue) /SUM(SUM(Revenue)) OVER (PARTITION BY 4-Months, StarRating, PaymentTerms, CountryG)

SUM(SUM(Revenue))OVER (PARTITION BY Year, StarRating, CountryG, PaymentTerms)

FROM BOOKINGS, GEO_GUEST, JUNK-PURCHASE, TIME, ROOM_TYPE, HOTEL

WHERE <join conditions>

AND AirConditioning = ‘True’ AND ‘Safe = ‘True’

GROUP BY 4-Months, HotelName, PaymentTerms, CountryG, StarRating, Year

Commento:

SELECT 4-Months, HotelName, PaymentTerms, CountryG,

SUM(Revenue)/COUNT(DISTINCT Month),

(SUM(Revenue)/~~**SUM(Num_Booked_Rooms))**~~/SUM(SUM(Revenue)) OVER (PARTITION BY 4-Months, PaymentTerms, CountryG, StarRating),

SUM(SUM(Revenue)) OVER (PARTITION BY PaymentTerms, CountryG, Year, StarRating)

FROM HOTEL H, ROOM-TYPE RT, TIME T, JUNK-PURCHASE JP, GUEST_GEO G, BOOKINGS B

WHERE B.HotelID = H.HotelID AND B.RTID = RT.RTID AND B.TimeID = T.TimeID AND B.IDJP = JP.IDJP AND B.GeoID = G.GeoID

AND AirConditioning = 'Yes' AND Safe = 'Yes'

GROUP BY 4-Months, HotelName, PaymentsTerms, CountryG, StarRating, Year

Domanda 17

Risposta corretta

Punteggio ottenuto 1,00

1 point (-15% penalty for an incorrect answer)

Notation

- Tn: Identifier of transaction n
- B(Tn): Begin of transaction Tn

su 1,00

- C(Tn): Commit of transactionTn
- A(Tn): Abort of transaction Tn
- CK: checkpoint
- U(ox), I(ox), D(ox): update, insert, and delete operations

The following sequence of operations within a log file is given:

B(T1) U1(o1) B(T2) I2(o2) CK(T1,T2) C(T2) B(T3) D3(o3) U3(o2) C(T3) U1(o4) FAILURE

What are the final UNDO and REDO sets for the warm restart procedure?

a. UNDO = {T1} REDO = {T2, T3} ✔

b. UNDO = {T4} REDO = {T1, T3}

c. None of the other answers is correct

d. UNDO = {T3} REDO = {T1, T2}

e. UNDO = {T2, T3, T4} REDO = {T1}

f. UNDO = {T1, T4} REDO = {T3}

Risposta corretta.

La risposta corretta è:
UNDO = {T1} REDO = {T2, T3}